

COURSE DESCRIPTION			
Program	Artificial Intelligence, Artificial Intelligence & Machine Learning, Architecture, Automobile Engineering, Biomedical Engineering, Cloud Computing and Big Data, Civil Engineering, Cyber Forensics and Information Security, Computer Science and Technology, Civil Engineering & Planning, Civil & Rural Engineering, Computer Science & Engineering, Computer Engineering, Civil & Environmental Engineering, Computer Science and Engineering (Artificial Intelligence & Machine Learning), Electronics and Communication, Electrical & Electronics Engineering, Electrical Engineering, Electronics Engineering, Electronics Engineering (Embedded Systems), Electronics and Computer Engineering, Electrical Engineering & Electric Vehicles Technology, Fire Technology and Safety, Food Processing Technology, Integrated Circuit Design & Fabrication, Interior Design, Instrumentation Engineering, Information Technology, Mechanical Engineering (Automobile Engg.), Mechatronics, Mechanical Engineering, Micro Electronics, Automation and Robotics, Renewable Energy, Robotic Process Automation, Civil Engineering (Construction Technology), Tool & Die Engineering, Wood and Paper Technology		
Course Title	Chemistry for Engineering Practices		
Course Code	2003A	Semester	I / II
Type of Course	Practicum	Contact Hours	90
Teaching Scheme (L: T:P:R)	3:0:3:0	Credits	4.5
CIE Marks	40	ESE Marks	60

Introduction

Chemistry for engineering practices is a foundational course designed to understand the basic concepts of Chemistry and their practical applications in engineering and daily life. This course enables students to develop problem-solving skills and gain knowledge necessary for selecting various engineering materials and addressing real-world industrial challenges.

Course Objectives

1	To impart knowledge about fundamental concepts in Chemistry including atoms, molecules, solutions, thermodynamics and relate them to engineering applications.
2	To apply key concepts of analytical chemistry and water treatment methods for solving real-world domestic and industrial problems.
3	To impart basic knowledge about electrochemistry and corrosion and enable students to solve problems in engineering systems.
4	To familiarize students with various engineering materials for selecting appropriate materials in industrial and domestic applications and understand green chemistry principles.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge of Chemistry			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply the basic concepts of atoms, molecules, solutions and thermodynamic processes in scientific and engineering contexts.	Apply
CO2	Apply the fundamentals of analytical Chemistry, appropriate water treatment methods and also the basic concepts of acids and bases to solve engineering problems.	Apply
CO3	Enable students to understand the basic knowledge of electrochemistry and corrosion and apply it in engineering systems to solve industrial challenges.	Apply
CO4	Explore various engineering materials for suitable industrial and domestic applications and explain green Chemistry principles promoting sustainable practices.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	-
CO2	3	3	2	3	2	2	2	-	-	-	-
CO3	2	2	2	2	2	-	-	-	-	-	-
CO4	3	2	3	2	2	2	3	-	-	-	-

Total	11	9	7	7	6	4	5	-	-	-	-
Correlation Level	2.67	2.25	2.3	2.3	2	2	2.5	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
3	0	3	0	6	90	4.5	20	60	80	20	0	20	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Fundamentals of Chemistry, Basic concepts of solution and Thermodynamics	Atoms and molecules: Fundamental particles- electrons, protons and neutrons), their charge and mass - Atomic number and mass number. Solutions: Solute, solvent and standard solution- methods to express the concentration of solution:- molarity, normality and ppm -Problems based on molarity and normality. Thermodynamics: System and surroundings - State variables (T, P &V)- Open, closed and isolated systems - Isothermal, isobaric, isochoric, adiabatic, reversible, irreversible, spontaneous, nonspontaneous, endothermic and exothermic processes - Internal energy, enthalpy and entropy.	11	0
Module I	Experiments on preparation of standard solutions & familiarization of endothermic and exothermic reactions	Preparation of 0.1 N oxalic acid solution, 0.1 N Sodium Carbonate solution, 0.1 M Zinc Sulphate solution. 0.1 M Magnesium Sulphate solution and demonstration of endothermic and exothermic reactions.	0	9
Module II	Basic concepts of Analytical Chemistry, Water treatment methods and Key concepts of acids and bases	Volumetric Analysis: Titration, end point, indicators and standard solution, choice of indicators in acid-base titration- Principle of Volumetric Analysis ($V_1N_1 = V_2N_2$), problems. Water: Soft and hard water - Salts causing water hardness - Temporary and permanent hardness. Water Treatment: Effluent treatment methods-Physical, chemical, biological and advanced technique (Flow chart) - Potable water and its characteristics- Municipal water treatment methods(in brief only): Screening, sedimentation, coagulation, filtration and sterilization:- Chlorination, UV radiation, and ozonation- Flow chart for municipal water treatment-Water filtration methods: Reverse osmosis and nanofiltration. Acids and bases (Arrhenius concept)- pH and pOH- pH equation- pH scale- Relation between pH and pOH - Simple problems based on pH - Applications of pH.	12	0
Module II	Quantitative & Qualitative Analysis	Standardization of HCl using standard sodium carbonate solution and standardization of EDTA using $ZnSO_4$. Estimation of sodium hydroxide using standard hydrochloric acid, potassium hydroxide solution using standard oxalic acid solution, total hardness of given water sample using standard EDTA solution. pH Determination using different techniques- pH meter, Universal indicator and pH test paper.	0	12
Module III	Electrochemistry, Corrosion and Emerging Battery Technologies	Electrochemistry: Oxidation, reduction and redox reactions - conductors and insulators - metallic conductors and electrolytic conductors- strong electrolytes, weak electrolytes and non-electrolytes- electrolysis, electrolytic cell- Faraday's laws of electrolysis -simple problems based on Faraday's laws -Industrial applications of electrolysis: electroplating and electrolytic refining - Electrochemical cell (Daniel cell)- classification: Primary cell, secondary cell and fuel cell. Corrosion of metals - Factors affecting rate of corrosion - Corrosion prevention methods: a) Barrier protection-metallic (anodic, cathodic) coatings, non-metallic (anodising and anti-rust solution) b) Cathodic protection- sacrificial anode method. Emerging battery technologies (Basic idea only) with the following examples: Li-ion, Na-ion, solid state battery.	11	0

Module III	Experiments based on electrochemical cell, conductivity and electrolysis	Construction of electrochemical cell, conductivity measurement and electroplating.	0	9
Module IV	Engineering Materials & Green Chemistry	Alloys: Purposes of alloying, applications of alloys like stainless steel, brass, bronze, and solder. Polymers: Monomers, polymers, polymerization, homo and copolymers, thermoplastics, and thermosetting plastics- common polymers: PE, PVC, Nylon-66, Bakelite, PET, PU, PBT, PC, and PMMA- Applications of each polymer- Natural rubber, vulcanization of rubber, and properties of vulcanized rubber- Synthetic rubber. Nanomaterials and nanotechnology- Carbon nanotubes, fullerenes, graphene- Applications of nano materials. Metal Organic Frameworks (MOFs) as advanced engineering materials - Applications. Biomaterials: Metallic, ceramic, polymeric and composites - Applications of biomaterials. Biofuel and its applications. Green Chemistry - Principles, importance and applications.	11	0
Module IV	Experiments based on estimation of metals in alloys and preparation of polymers	Estimation of metals in alloys and preparation of polymers.	0	9

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Atoms and molecules	Define atom and molecule with examples.	L	CO1	Remember	0.5
Atoms and molecules	List the fundamental particles of atom (electrons, protons and neutrons) and give their charge and mass.	L	CO1	Understand	0.5
Atoms and molecules	Define the terms atomic number and mass number with examples.	L	CO1	Remember	0.5
Atoms and molecules	Calculate the number of fundamental particles of an atom using atomic number and mass number of elements (upto atomic number Z=20).	L	CO1	Apply	0.5
Solutions	Define the terms solute, solvent and standard solution with examples.	L	CO1	Remember	1
Solutions	Define the terms molarity, normality and ppm and give the equations for molarity, normality and ppm.	L	CO1	Understand	1
Solutions	Solve problems based on molarity and normality.	L	CO1	Apply	2
Thermodynamics-	Define the terms system and surroundings in thermodynamics and state variables (T, P & V).	L	CO1	Remember	1
Thermodynamics-	Explain different types of systems- open, closed and isolated systems with example.	L	CO1	Understand	1
Thermodynamics-	Explain thermodynamic processes- isothermal, isobaric, isochoric and adiabatic with example.	L	CO1	Understand	0.5
Thermodynamics-	Differentiate between reversible and irreversible processes with example.	L	CO1	Understand	0.5
Thermodynamics-	Differentiate between spontaneous and non spontaneous processes with example.	L	CO1	Understand	0.5
Thermodynamics-	Differentiate between endothermic and exothermic processes with example.	L	CO1	Understand	0.5
Thermodynamics-	Define the terms internal energy, enthalpy and entropy.	L	CO1	Remember	1
Preparation of standard solutions and demonstration of endothermic and exothermic reactions.	Prepare 0.1 N oxalic acid solution	P	CO1	Apply	1.5
Preparation of standard solutions and demonstration of endothermic and exothermic reactions.	Prepare 0.1 N sodium carbonate solution	P	CO1	Apply	1.5

Preparation of standard solutions and demonstration of endothermic and exothermic reactions.	Prepare 0.1 M Zinc Sulphate solution	P	CO1	Apply	1.5
Preparation of standard solutions and demonstration of endothermic and exothermic reactions.	Prepare 0.1 M Magnesium Sulphate solution	P	CO1	Apply	1.5
Preparation of standard solutions and demonstration of endothermic and exothermic reactions.	Demonstrate endothermic and exothermic reactions (Endothermic: The reaction of citric acid and baking soda, the reaction of barium hydroxide and ammonium chloride; Exothermic: Neutralisation between acids and bases- (e.g., HCl and NaOH reaction,) and reaction of NaOH in water, reaction of CaCl ₂ in water.	P	CO1	Apply	3
Module II					
Volumetric analysis	Define the terms titration, end point, indicator and standard solution in volumetric analysis.	L	CO2	Remember	0.5
Volumetric analysis	Explain the choice of indicator in the following titrations: Strong acid- strong base, strong acid- weak base, weak acid-strong base(with example).	L	CO2	Understand	0.5
Volumetric analysis	State the Principle of Volumetric Analysis using normality equation ($V_1N_1 = V_2N_2$).	L	CO2	Remember	0.5
Volumetric analysis	Solve problems based on normality equation $V_1N_1 = V_2N_2$.	L	CO2	Apply	1
Hardness of water	Differentiate between soft and hard water. List the salts causing hardness of water. Differentiate between temporary and permanent hardness of water.	L	CO2	Understand	1
Water treatment methods	Discuss the different types of effluent treatment methods: Physical, chemical, biological and advanced technique (in brief only).	L	CO2	Understand	1
Water treatment methods	Draw the flow chart for the effluent treatment method.	L	CO2	Remember	0.5
Water treatment methods	Define potable water and list its characteristics.	L	CO2	Remember	0.5
Water treatment methods	Discuss the different steps involved in the municipal water treatment and explain each step briefly (Screening, sedimentation, coagulation, filtration and sterilization).	L	CO2	Understand	2
Water treatment methods	Explain different sterilization methods of water: Chlorination, UV radiation, and ozonation.	L	CO2	Understand	0.5
Water treatment methods	Draw the flow chart for municipal supply of potable water.	L	CO2	Remember	0.5
Water treatment methods	Explain different filtration methods of water (reverse osmosis and nanofiltration).	L	CO2	Understand	0.5
Acids ,bases - pH and POH	Define acids and bases according to the Arrhenius concept with examples. Define the terms pH and pOH. State the relation between pH and pOH. Define pH scale.	L	CO2	Understand	1
Acids ,bases - pH and POH	Solve problems based on pH.	L	CO2	Apply	1.5
Acids ,bases - pH and POH	Discuss the applications of pH.	L	CO2	Understand	0.5
Quantitative and Qualitative analysis	Standardisation of hydrochloric acid using standard sodium carbonate solution.	P	CO2	Apply	3
Quantitative and Qualitative analysis	Estimation of sodium hydroxide using standard hydrochloric acid.	P	CO2	Apply	3
Quantitative and Qualitative analysis	Estimation of potassium hydroxide solution by titrating against standard oxalic acid solution.	P	CO2	Apply	3
Quantitative and Qualitative analysis	Standardisation of EDTA using ZnSO ₄ .	P	CO2	Apply	3
Quantitative and Qualitative analysis	Volumetric estimation of total hardness of given water sample using standard EDTA solution.	P	CO2	Apply	3

Quantitative and Qualitative analysis	Determine the pH of solutions using various techniques- pH meter, Universal indicator and pH test paper.	P	CO2	Apply	3
Module III					
Electrochemistry	Define oxidation, reduction, redox reactions, conductors and insulators with suitable examples for each.	L	CO3	Remember	1
Electrochemistry	Differentiate between metallic conductors (electronic conductors) and electrolytic conductors (ionic conductors) with examples.	L	CO3	Understand	1
Electrochemistry	Define electrolytes (strong and weak electrolytes), non-electrolytes, electrolysis and electrolytic cell with suitable examples.	L	CO3	Remember	1
Electrochemistry	State Faraday's laws of electrolysis (First and second law).	L	CO3	Remember	1
Electrochemistry	Solve problems based on Faraday's laws (First and second law).	L	CO3	Apply	1
Electrochemistry	Illustrate industrial applications of electrolysis—electroplating (electroplating of Nickel on mild steel) and electrolytic refining (electrolytic refining of copper).	L	CO3	Understand	1
Electrochemistry	Illustrate electrochemical cell using Daniel cell as example and also explain salt bridge.	L	CO3	Understand	1
Electrochemistry	Classify electrochemical cell into primary cell, secondary cell and fuel cell with one example for each.	L	CO3	Understand	1
Corrosion	Define corrosion of metals and list factors affecting rate of corrosion of metals.	L	CO3	Remember	0.5
Corrosion	Explain corrosion prevention methods: a) Barrier protection-metallic (anodic, cathodic) coatings, non-metallic (anodising and anti rust solution) b) Cathodic protection- sacrificial anode method.	L	CO3	Understand	2
Emerging battery technologies	Discuss about emerging battery technologies (Basic idea only) with the following examples: Li-ion, Na-ion, solid state battery.	L	CO3	Understand	0.5
Experiments based on electrochemical cell conductivity and electrolysis	Construct salt bridge	P	CO3	Apply	1.5
Experiments based on electrochemical cell conductivity and electrolysis	Construct Daniel cell	P	CO3	Apply	1.5
Experiments based on electrochemical cell conductivity and electrolysis	Electroplating—Coating a steel spoon with Nickel (mixture of NiCl ₂ and NiSO ₄ as electrolyte)	P	CO3	Apply	3
Experiments based on electrochemical cell conductivity and electrolysis	Demonstrate the conductivity of a given water sample using a conductivity meter	P	CO3	Apply	1.5
Experiments based on electrochemical cell conductivity and electrolysis	Demonstrate electrolysis of water	P	CO3	Apply	1.5
Module IV					
Alloys	Define an alloy with examples. List the purposes of alloying. List any two applications of alloys like stainless steel, brass, bronze and solder.	L	CO4	Remember	1
Polymers	Define monomers, polymers and polymerization with examples and also differentiate between homo and copolymers - with one example for each.	L	CO4	Understand	1
Polymers	Differentiate between thermoplastics and thermosetting plastics- with one example for each.	L	CO4	Understand	1
Polymers	Discuss the monomers and any two applications of common polymers - PE, PVC, Nylon-66, Bakelite, PET, PU, PBT, PC, PMMA.	L	CO4	Understand	2

Polymers	Name the monomer of natural rubber and explain the process of vulcanization of rubber and also list the properties of vulcanized rubber.	L	CO4	Understand	1
Polymers	Define synthetic rubber and List the monomers of synthetic rubber—Buna-S and Buna-N	L	CO4	Remember	1
Nanomaterials	Define nanomaterials and nanotechnology. Discuss the nanomaterials: Carbon nanotubes, fullerenes, and graphene(basic idea only) List the applications of nanomaterials.	L	CO4	Understand	1
Metal Organic Frameworks (MOFs)	Define Metal-Organic Frameworks (MOFs) and list any three applications.	L	CO4	Remember	1
Biomaterials	Define biomaterials and list any three applications. Discuss briefly about biomaterials (metallic, ceramic, polymeric, and composite). Define biofuel with examples and discuss the applications of biofuel.	L	CO4	Understand	1
Green Chemistry	Discuss the principles of Green Chemistry and also discuss the importance and applications of green chemistry.	L	CO4	Understand	1
Experiments are based on estimation of metals in alloys and preparation of polymers	Estimation of copper in brass.	P	CO4	Apply	3
Experiments are based on estimation of metals in alloys and preparation of polymers	Estimation of zinc in brass.	P	CO4	Apply	3
Experiments are based on estimation of metals in alloys and preparation of polymers	Estimation of iron in iron ore	P	CO4	Apply	3
Experiments are based on estimation of metals in alloys and preparation of polymers	Preparation of: Urea-formaldehyde resin	P	CO4	Apply	3
Experiments are based on estimation of metals in alloys and preparation of polymers	Preparation of Phenol-formaldehyde resin	P	CO4	Apply	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Prepare an assignment to find the atomic number and mass number of any 20 elements. Draw the structure of any 5 atoms and label the nucleus and fundamental particles. Find the atoms and molecules present in 5 substances around you. Prepare a simple model of atom using chart/ coloured paper/ seeds etc. Completion of record work and result analysis.	6
2	Write a note on concentration, methods of expressing concentration terms. Find the normality and molarity of different solutions(simple problems) . Find 5-10 solutions available at home and prepare a short report including: Name of solution, Solute, Solvent etc. Completion of record work and result analysis.	6
3	Write a report on the different thermodynamic processes and identify daily life examples in detail. Observe endothermic and exothermic processes at home and make a report. Completion of record work and result analysis.	6
4	Find any 4 spontaneous and non-spontaneous processes in everyday life. Record the principle and procedure for the preparation of standard solutions. Completion of record work and result analysis.	6
Module II		
5	Assignment on extra problems based on normality equation ($N_1V_1=N_2V_2$) Make a study report on choice of indicators- Selection of suitable indicators for different types of titrations. Identify and prepare simple natural indicators using household materials. Completion of record work and result analysis.	6
6	Collect water samples from different sources and other solutions and identify pH. Identify soft water vs hard water through daily-life observation. Make a report on problems caused by hard water at home. Completion of record work and result analysis.	6

7	Identify different methods involved in modern water filters. Write a report on major stages in municipal water treatment and draw the flow chart. Do the sedimentation and filtration using muddy water. Completion of record work and result analysis.	6
8	Prepare a report on water purification methods in your locality. Prepare for an assignment on CO1 and CO2 Prepare for lab series - 1 exam Completion of record work and result analysis.	6
Module III		
9	Write a report on hydrogen economy (use of hydrogen fuel in automobiles) Completion of record work and result analysis.	6
10	Prepare an assignment on various oxidation-reduction reactions and explain the working of Daniel cell with diagram Completion of record work and result analysis.	6
11	Write a report on recycling of batteries Completion of record work and result analysis.	6
12	Prepare a report on types of batteries used in electric vehicles and also explain the advantages and disadvantages of electric vehicle batteries Completion of record work and result analysis.	6
Module IV		
13	Prepare an assignment on articles made of alloys used in daily life. Write a note on bioplastics/biodegradable polymers. Present a report of the polymeric materials which can be reused and recycled. Prepare a short video of major nanomaterials and its applications. Completion of record work and result analysis.	6
14	Find the materials used for biomedical implants. Prepare an assignment on biomaterials. Prepare a report on the applications of metal organic frameworks. Completion of record work and result analysis.	6
15	Identify the procedure for any green synthesis. Prepare an assignment on biofuels. Completion of record work and result analysis. Prepare for an assignment on CO3 and CO4 Prepare for lab series - 2 exam.	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Principles of Physical Chemistry	Puri, Sharma & Pathania	Vishal Publishing.Co.
2	Textbook of Physical Chemistry	P.L.Soni, O.P.Dharmarha , U.N.Dash	
3	Vogel's Textbook of Quantitative Chemical Analysis	Arthur Israel Vogel	Prentice Hall, 2000
4	Fundamentals of Analytical Chemistry	A Douglas, Skoog and Donald M. West, F.J. Holler	8thEdn., Harcourt College Publishers, 2004
5	The Essentials: Understanding Nanoscience and Nanotechnology;	Pradeep, T.	1st Edition ed.; McGraw-Hill Education: New York, 2007
6	Graphene Synthesis, Properties,Technology and Applications	Saritha Appukkutan,Jitha S. Jayan, Sethulakshmi A.S	Woodhead Publishing
7	Nanoscience and Technology	Muralidharan, V. S.; Subramania	New Delhi : Ane Books for ; Boca Raton : CRC Press
8	Practical Chemistry	A.O. Thomas	Scientific Book Centre, Cannanore / Kannur
9	Water and waste water engineering	Sudha Goel	Cambridge University Press

REFERENCE			
Sl. No.	Title	Author	Publication

1	Physical Chemistry	Atkins	Oxford University Press (OUP)
2	Textbook of Physical Chemistry	P.L.Soni, O.P.Dharmatha, Dash U N	Sulthan Chand & Sons

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in/courses/104106119	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Conductivity meter	LED/LCD display range upto 1000mS/cm(230V)	1
2	pH meter	Digital/benchtop/portable/pen type	1
3	Bunsen burner		2
4	Digital weighing balance		1
5	Thermometer		1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	
Exam mark		15	50	40	40	50	50	60	
Converted to		5	10	10	10	10	10		
Marks	5	5	10	10		10		60	
Total	40							60	
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

Note:

Experiments shall be conducted such that all COs are attained.

Minimum of 2 experiments from each module shall be performed and total 10 experiments from all modules should be performed.